

Save the Arduino code as a .ino file and provide it as a downloadable link

```
arduino_code = ""
```

```
#include <Wire.h>
```

```
#include "DHT.h"
```

```
#define DHTPIN 33
```

```
#define DHTTYPE DHT11
```

```
#include <WiFi.h>
```

```
#include <PubSubClient.h>
```

```
DHT dht(DHTPIN, DHTTYPE);
```

```
const char *ssid = "Football";
```

```
const char *password = "football2004";
```

```
const char *mqtt_broker = "broker.emqx.io";
```

```
const char *topic = "esp32/test";
```

```
const char *mqtt_username = "emqx";
```

```
const char *mqtt_password = "public";
```

```
const int mqtt_port = 1883;
```

```
WiFiClient espClient;
```

```
PubSubClient client(espClient);
```

```
const int mq2 = 32;
```

```
const int mq4 = 34;
```

```
const int mq135 = 35;
```

```
void setup() {
```

```
Serial.begin(9600);

dht.begin();

pinMode(mq4, INPUT);
pinMode(mq135, INPUT);
pinMode(mq2, INPUT);


WiFi.begin(ssid, password);
while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}
Serial.println("");
Serial.println("WiFi connected");
Serial.println("IP address: ");
Serial.println(WiFi.localIP());


mqttConnect();
}


void loop() {
    client.loop();

    float h = dht.readHumidity();
    float t = dht.readTemperature();
    float f = dht.readTemperature(true);

    if (isnan(h) || isnan(t) || isnan(f)) {
        Serial.println(F("Failed to read from DHT sensor!"));
        return;
    }

    float hif = dht.computeHeatIndex(f, h);
```

```
float hic = dht.computeHeatIndex(t, h, false);
```

```
double mq4_value = analogToDigital(analogRead(mq4));
```

```
double mq135_value = analogToDigital(analogRead(mq135));
```

```
double mq2_value = analogToDigital(analogRead(mq2));
```

```
char output[15];
```

```
dtostrf(mq4_value, 8, 4, output);
```

```
client.publish("MQ4", output);
```

```
dtostrf(mq2_value, 8, 4, output);
```

```
client.publish("MQ2", output);
```

```
dtostrf(mq135_value, 8, 4, output);
```

```
client.publish("MQ135", output);
```

```
dtostrf(f, 8, 4, output);
```

```
client.publish("DHT-temp", output);
```

```
dtostrf(h, 8, 4, output);
```

```
client.publish("DHT-Humidity", output);
```

```
Serial.println("*****");
```

```
Serial.print("MQ-4 Value: "); Serial.print(mq4_value, DEC); Serial.println(" PPM");
```

```
Serial.print("MQ-135 Value: "); Serial.print(mq135_value, DEC); Serial.println(" PPM");
```

```
Serial.print("MQ-2 Value: "); Serial.print(mq2_value, DEC); Serial.println(" PPM");
```

```
Serial.print("DHT Temperature: "); Serial.print(f); Serial.println(" °C");
```

```
Serial.print("DHT Humidity: "); Serial.print(h); Serial.println(" %");
```

```
Serial.println("*****\\n\\n");
```

```
delay(3000);
```

```
}
```

```
double analogToDigital(int aValue) {
```

```
    Serial.println(aValue);
```

```
float sensor_volt = aValue * (3.3 / 4095.0);  
return sensor_volt;  
}
```

```
void mqttConnect() {  
    client.setServer(mqtt_broker, mqtt_port);  
    client.setCallback(callback);  
    while (!client.connected()) {  
        String client_id = "esp32-client-";  
        client_id += String(WiFi.macAddress());  
        Serial.printf("The client %s connects to the public mqtt broker\\n", client_id.c_str());  
        if (client.connect(client_id.c_str(), mqtt_username, mqtt_password)) {  
            Serial.println("Public emqx mqtt broker connected");  
        } else {  
            Serial.print("failed with state ");  
            Serial.print(client.state());  
            delay(2000);  
        }  
    }  
    client.publish(topic, "Hi EMQX I'm ESP32 ^^");  
    client.subscribe(topic);  
}
```

```
void callback(char *topic, byte *payload, unsigned int length) {  
    Serial.print("Message arrived in topic: ");  
    Serial.println(topic);  
    Serial.print("Message:");  
    for (int i = 0; i < length; i++) {  
        Serial.print((char) payload[i]);  
    }  
    Serial.println();  
}
```

```
Serial.println("-----");  
}  
"""
```

```
file_path = "/mnt/data/loT_Environment_Monitoring_System.ino"  
with open(file_path, "w") as f:  
    f.write(arduino_code)
```

```
file_path
```